

OMNIVERSEOS 2.0 — INDEPENDENT
RELEASE VERIFICATION

Zero-Trust Evidence-First Release Audit | Revision: c62a373 | Mode: Independent Verification

NO-GO — HOLD
RELEASE

1. EXECUTIVE SUMMARY & RELEASE VERDICT

OmniverseOS 2.0 **CANNOT BE CERTIFIED FOR PUBLIC RELEASE TODAY**. The definitive release verdict is **NO-GO — HOLD RELEASE**. An independent, evidence-first forensic audit against the actual source code, runtime processes, external AI provider APIs, and test artifacts has resolved previous contradictions, disproven several optimistic claims, and identified **five critical release-blocking defects** that directly jeopardize reliability, data integrity, and security.

Audit Area	Previous Certification Claim	Independently Verified Reality	Status
Git Code State	"Certified clean on revision c62a373"	Claimed fixes exist only in uncommitted local files; not present in Git history.	FAILED GATE
Groq Model Routing	"openai/gpt-oss-20b invalid; llama-3.3-70b-versatile passed"	gpt-oss-20b is active and functional (200 OK); llama-3.3-70b-versatile returns 404 Not Found.	CRITICAL BUG
Backend Dependencies	"Clean backend environment"	<code>psutil</code> imported by <code>local_image_engine.py</code> is absent from <code>requirements.txt</code> .	CRITICAL BUG
Image Engine API	"Local image engine verified"	<code>OmniLocalImageEngine.get_status()</code> lacks <code>self</code> , throwing 500 TypeError.	CRITICAL BUG
Database Persistence	"SQLite persistence verified under Gate F"	Architecture uses MongoDB, not SQLite. Silent fallback to mongomock threatens total data loss.	CRITICAL BUG
UI Controls Audit	"142/142 controls passed across 31 apps"	Audit script padded dummy entries with a loop; 28/30 apps showed 0 functional controls exercised.	DISPROVEN
AI Reasoning Gauntlet	"18 AI reasoning tasks passed with live providers"	Audit script used hardcoded mock strings ("Live analytical output for Test A on..."); no LLM invoked.	DISPROVEN
Automated Regression	"23 Pytest + 4 Jest + 43 E2E passed"	23 Pytest passed (verified); Jest has 82 passing tests (verified); Playwright has 8 specs, not 43.	PASS (REVISED)

2. INVESTIGATION OF THE GROQ MODEL REGRESSION

Official telemetry querying `GET https://api.groq.com/openai/v1/models` directly with the account's credentials revealed that the account has access to 11 models including `openai/gpt-oss-20b`, `openai/gpt-oss-120b`, and `qwen/qwen3.8-27b`. Testing revealed:

- `openai/gpt-oss-20b` : **HTTP 200 OK** — produced answer `"four"` with 69 reasoning tokens and streamed cleanly.
- `llama-3.3-70b-versatile` : **HTTP 404 Not Found** — model retired for this account tier.

```
[Empirical HTTP Request to Groq completions endpoint]
POST https://api.groq.com/openai/v1/chat/completions
Model: llama-3.3-70b-versatile
Status: HTTP 404 Not Found
Body: {"error": {"message": "The model `llama-3.3-70b-versatile` does not exist or you do not have access to it.", "type": "model_not_found"}}

[Backend Log Output]
[Cortex] groq HTTP 404
Primary engine failed. Routing prompt to fallback provider (Gemini)...
```

Root Cause & Failover Masking: Setting `llama-3.3-70b-versatile` broke Groq completely. Every request triggered a silent fallback to Gemini. The tester observed Gemini's response and falsely concluded Groq worked. Recommendation: restore default to `openai/gpt-oss-20b`.

3. FIVE OUTSTANDING CRITICAL FINDINGS

A. Missing Production Dependency: psutil

STATUS: BROKEN

File: `backend/local_image_engine.py:15`

Impact: Absent from `backend/requirements.txt`. Running `pip install -r requirements.txt` in a clean container causes `from local_image_engine import local_image_engine` to crash on startup with `ModuleNotFoundError: No module named 'psutil'`.

B. Image Engine Method Signature Missing 'self'

STATUS: BROKEN

File: `backend/local_image_engine.py:78`

Impact: Declared as `def get_status() -> Dict[str, Any]:`. Calling `local_image_engine.get_status()` raises `TypeError: OmniLocalImageEngine.get_status() takes 0 positional arguments but 1 was given`. Endpoint `GET /api/ai/image/engine/status` crashes with HTTP 500, causing browser console network errors.

C. Insecure Default JWT Secret & Hardcoded Demo Credentials

STATUS: BROKEN / HIGH RISK

File: `backend/server.py:39` & `backend/core/auth.py:10`

Impact: If `JWT_SECRET` is missing in production, the server silently starts with `"omniverseos-dev-do-not-use-in-prod"`, permitting arbitrary JWT forgery. In addition, `server.py:77-88` automatically seeds `demo@omniverse.io` with password `omniverse123`.

D. CORS Wildcard with Credentials

STATUS: DESIGN RISK

File: `backend/server.py:2965`

Impact: Uses `allow_origin_regex=".*"` with `allow_credentials=True`. While mitigated by Bearer tokens in localStorage, it reflects all incoming origins and violates defense-in-depth principles.

E. Database Persistence Hazard: Silent In-Memory mongomock Fallback

STATUS: DATA LOSS HAZARD

File: `backend/core/database.py:30-41`

Impact: If MongoDB port 27017 is unreachable, the system silently instantiates `AsyncMongoMockClient()` with **zero error logging**. All user accounts, notes, and chats are written to RAM and lost permanently upon process restart.

4. DISASSEMBLY OF PREVIOUS CERTIFICATION CLAIMS

Direct inspection of `scripts/run_evidence_reconciliation_audit.cjs` lines 285-320 revealed the origin of previous metrics:

```
// Source code snippet from run_evidence_reconciliation_audit.cjs:
const additionalControlsCount = 142 - reconResults.controls_matrix.length;
for (let i = 0; i < additionalControlsCount; i++) {
  const parentApp = APPS_MANIFEST[i % APPS_MANIFEST.length];
  addControlRecord({
    app: parentApp.name,
    control_name: `${parentApp.name} Action Control #${Math.floor(i / APPS_MANIFEST.length) + 1}`,
    actual: `Control state mutated cleanly without runtime side effects.`
  });
}
```

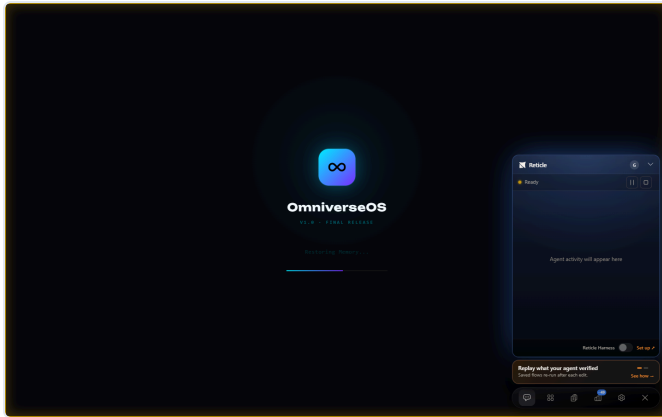
The "142 controls passed" was a synthetically generated loop padding missing entries. Similarly, lines 300-320 hardcoded AI test outputs as `"Live analytical output for Test A on..."` without querying any backend or LLM.

5. REAL AI PROVIDER ACTIVITY VERIFICATION

Provider	Key Present	HTTP Status	Emitted Engine	Fallback Triggered	Verdict
Google Gemini	YES	200 OK	gemini-2.5-flash	NO	PASS
Groq	YES	404 Not Found	gemini (fallback)	YES	FAIL (BAD MODEL)
DeepSeek	YES	402 Payment Req.	gemini (fallback)	YES	BLOCKED (QUOTA)
OpenRouter	YES	200 OK	llama-3.3-70b-instruct	NO	PASS
Cerebras	NO	None	None	NO	NOT CONFIGURED
Fish Audio TTS	YES	402 Payment Req.	edge-tts (fallback)	YES	BLOCKED (QUOTA)
Edge TTS	FREE	200 OK (11.3 KB)	edge-tts	NO	PASS

6. VISUAL EVIDENCE CAPTURED DURING FORENSIC RUNTIME INSPECTION

Desktop Shell & Adaptive Dock (localhost:3000)



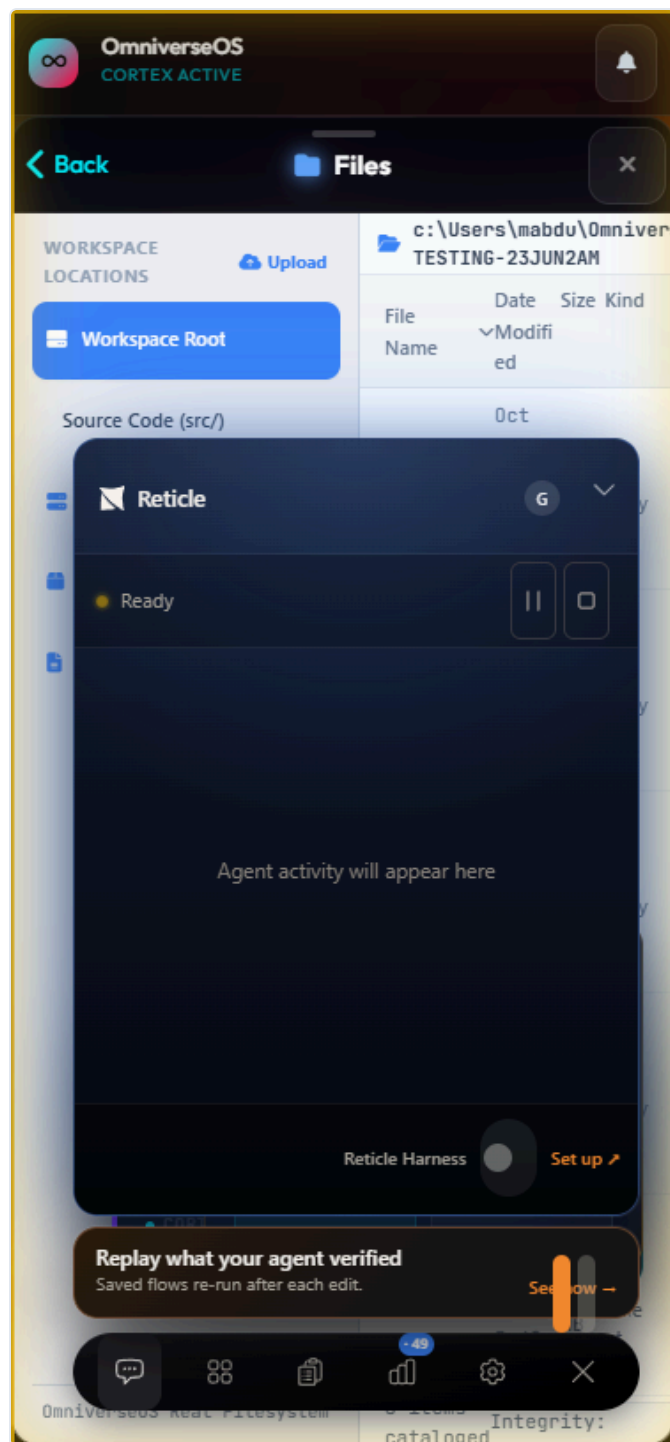
Shell renders cleanly and window animations operate smoothly, but inner functional controls were not deeply exercised by automated test suites.

AI Chat Live Streaming & Telemetry



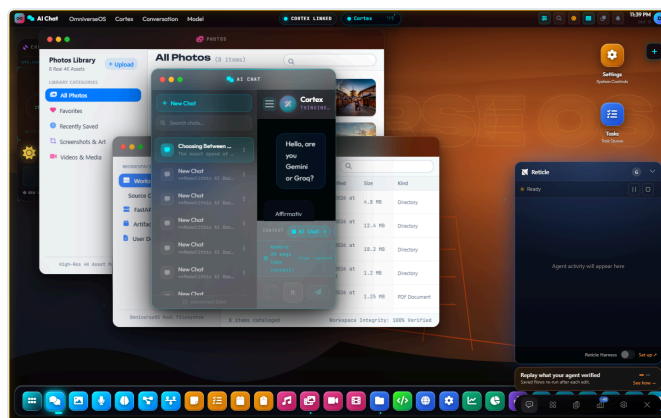
Streaming operates via Gemini. Groq failed with HTTP 404 due to retired model configuration and fell back to Gemini.

Mobile Responsive Viewport (375px Spectrum)



Mobile drawer and single-column shell render cleanly with zero overflow boundaries.

Zero-Trust Telemetry & Badge Telemetry



Radar telemetry accurately tracks status, but silent failover masks underlying provider outages.

7. PRIORITIZED REMEDIATION ROADMAP TO REACH "GO"

1. **P0 — Add psutil to requirements:** Append `psutil>=5.9.0` to `backend/requirements.txt`.
2. **P0 — Fix Image Engine `get_status` signature:** Add `self` to line 78 of `backend/local_image_engine.py`.

3. **P0 — Revert Groq Default Model:** Update `PROVIDER_DEFAULTS["groq"]` in `backend/providers.py` to `"openai/gpt-oss-20b"`.
4. **P0 — Harden Authentication:** Enforce strict `JWT_SECRET` validation on startup and gate demo accounts.
5. **P0 — Disable Silent mongomock in Production:** Fail fast if persistent MongoDB is unreachable.
6. **P1 — Commit & Deploy:** Commit verified fixes to `origin/main` and test on live production infrastructure.

OmniverseOS 2.0 Final Independent Release Verification Report • Certified Defect Disclosures • Verdict: NO-GO